MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSECFS202 – PARALLEL PROCESSING

Time:2 Hours MaximumMarks:70

Part A

Short Answer Type Questions
Answer any **10** Questions
Each question carries **2** marks

1. Summarize parallel processing. [Understand][1]
2. Identify the types of architectural configuration of parallel computers [Understand][1]
3. Compare serial vs parallel processing. [Understand][1]
4. What is linear Pipelining. [Analyse][2]
5. Predict any four classifications of pipeline processor. [Analyse][2]
6. Simplify register tagging. [Analyze][2]
7. Examine the characteristics of vector processing. [Analyse][2]
8. Outline the pipelined vector processing method. [Analyse][2]
9. Compare static and dynamic network. [Understand][3]
10. Categorize the classification of multiprocessor operating system. [Understand][4]
11. Outline the process synchronization mechanism. [Understand][4]
12. Distinguish between Control flow and data flow computers. [Understand][4]

Part B

Short Essay Type Questions
Answer any **6** Questions
Each question carries **5** marks

13. Explain the types of generations of computer systems. [Understand][1]
14. Describe Parallel processing mechanism. [Understand][1]
15. Differentiate SIMD, MISD and MIMD. [Understand][1]
16. Discuss the Parallel processing application. [Understand][1]
17. Illustrate arithmetic pipeline design examples. [Analyse][2]
18. Explain masking and data routing mechanism. [Understand][3]
19. Distinguish between the loosely coupled and the tightly coupled multiprocessor. [Understand][4]

20. Simplify crossbar switch and multiport memories. [Understand][4]

Part C

Essay Type Questions

Answer any **2** Questions

Each question carries **10** marks

21. Describe architectural classification scheme. [Understand][1]

22. Illustrate the principles of designing pipeline processor. [Analyse][2]

23. Explain parallel algorithms for array processors. [Understand][3]